5. When I refer to an object's mass in this context, I am referring to the sum total mass of its particulate constituents. If a cube, say, were composed of 1,000 gold atoms, I'd be referring to 1,000 times the mass of a single such atom. This definition jibes with Newton's perspective. Newton's laws say that such a cube would have a mass that is 1,000 times that of a single gold atom, and that it would weigh 1,000 times as much as a single gold atom. According to Einstein, though, the weight of the cube also depends on the kinetic energy of the atoms (as well as all other contributions to the energy of the cube). This follows from $E=mc^2$: more energy (E), regardless of the source, translates into more mass (m). Thus, an equivalent way of expressing the point is that because Newton didn't know about $E=mc^2$, his law of gravity uses a definition of mass that misses various contributions to energy, such as energy associated with motion.

6. The discussion here is suggestive of the underlying physics but does not capture it fully. The pressure exerted by the compressed spring does indeed influence how strongly the box is pulled earthward. But this is because the compressed spring affects the total energy of the box and, as discussed in the previous paragraph, according to general relativity, the total energy is what's relevant. However, the point I'm explaining here is that pressure itself—not just through the contribution it makes to total energy—generates gravity, much as mass and energy do. According to general relativity, pressure gravitates. Also note that the repulsive gravity we are referring to is the *internal* gravitational field experienced within a region of space suffused by something that has negative rather than positive pressure. In such a situation, negative pressure will contribute a repulsive gravitational field acting *within* the region.

7. Mathematically, the cosmological constant is represented by a number, usually denoted by Λ (see note 4). Einstein found that his equations made perfect sense regardless of whether Λ was chosen to be a positive or a negative number. The discussion in the text focuses on the case of particular interest to modern cosmology (and modern observations, as will be discussed) in which Λ is positive, since this gives rise to negative pressure and repulsive gravity. A negative value for Λ yields ordinary attractive gravity. Note, too, that since the pressure exerted by the cosmological constant is uniform, this pressure does not